

Section B

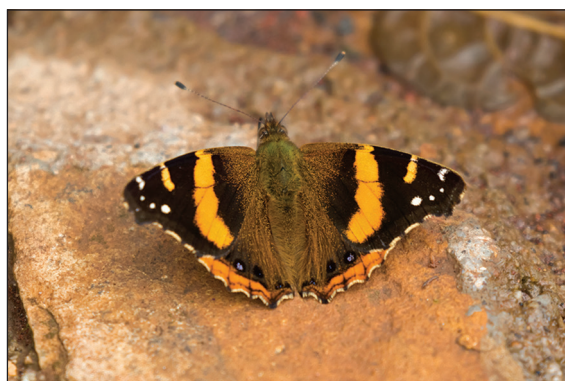
Refer to the separate Resource Folder to answer question 8.

8. (a) (i) The Red Admiral butterfly competes with greenfly for food. The scientific name for one type of greenfly is *Acyrtosiphon pisum*.

Complete the table below to show the scientific classification of greenfly. [3]

Kingdom:	
Phylum:	
Class:	Insecta
Order:	Hemiptera
Family:	Aphididae
Genus:	
Species:	
Scientific name:	<i>Acyrtosiphon pisum</i>

- (ii) Another butterfly is found in Africa. Its scientific name is *Vanessa abyssinica*.



- I. State **one** advantage of using scientific names. [1]

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- II. State **one** piece of evidence, from its scientific name, that shows it is related to the Red Admiral. [1]

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(b) The variation in population of adult Red Admirals is shown by the graph in **Figure 1**.

- (i) I. State the name of the seasonal movement of animals as seen in Red Admirals. [1]

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II. State **one** reason why this seasonal movement occurs. [1]

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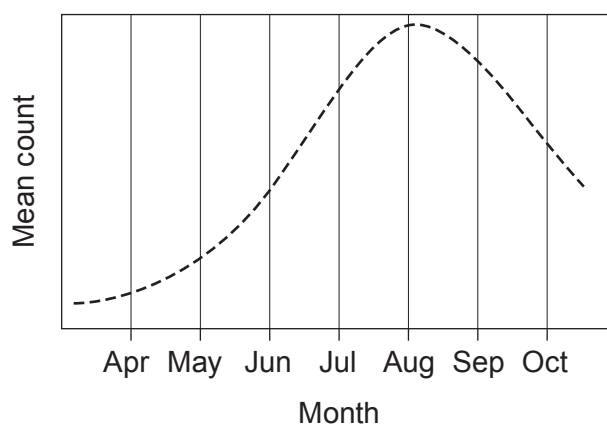
- (ii) Use the information in the graph in **Figure 1** to compare the variation of the Red Admiral population in 2019 with the mean change since 1976. [2]

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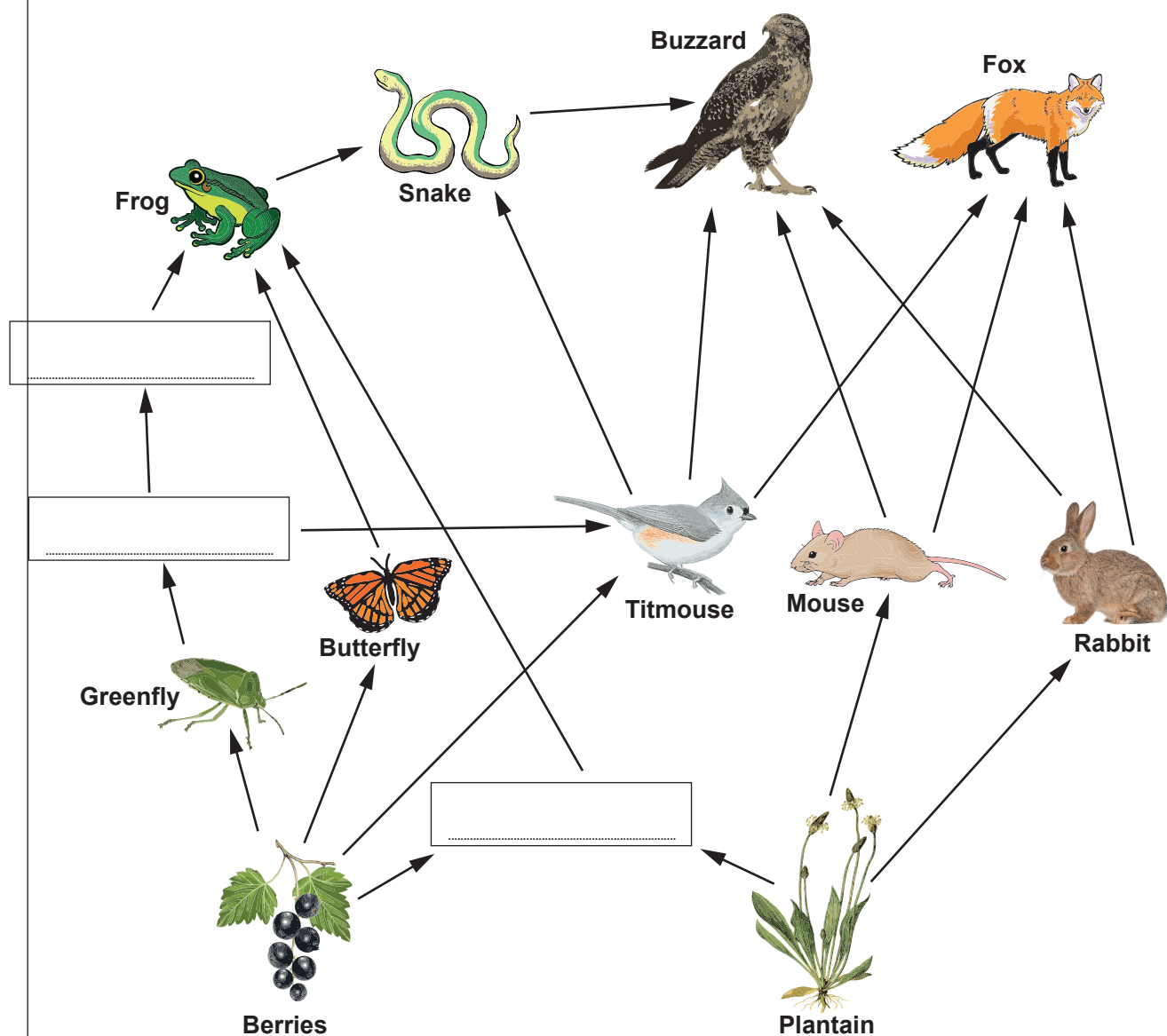
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- (iii) Red Admirals are a food source for frogs. **Add a line** to the graph below to show how the population of frogs changed between April and October in 2019. [2]



(c) **Table 2** gives information about the feeding relationships in one woodland habitat.

(i) Use the information in **Table 2** to **complete** the food web below. [3]



(ii) In this habitat, the fields containing plantain are cleared to make way for housing. Explain how this affects the foxes. [3]

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(iii) Explain why the frog can be found at more than one trophic level. [2]

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(d) Refer to **Table 2** to construct a pyramid of biomass for a food chain of **five** trophic levels starting with berries and ending with buzzards. [3]

(e) Use the information in **Figure 2** to calculate the efficiency of energy transfer between the primary and secondary consumers. [3]

% efficiency =

END OF PAPER

8





THE PERIODIC TABLE

Group 1 2 3 4 5 6 7 0

1 H Hydrogen 1 4 He Helium 2																	

Key

relative atomic mass		atomic number	
A_r	Symbol	Name	Z